

Data Quality Plan (DQP)

Dataset: geocoded_train.csv -> train-clean-final.csv

Module: COMP47350 Group Project DQP

Scope: Feature-level issue resolution, justification, final feature typing, and cleaned CSV export.

1. Executive Summary

This DQP defines the cleaning policy implemented in Part 2 of project.ipynb. The goal is to convert the geocoded training file into a modelling-ready dataset with fixed feature types, explicit rule-based cleaning, and no missing values.

The DQP starts from geocoded_train.csv with 54,000 rows and 14 columns. It ends with train-clean-final.csv containing 53,991 rows and 10 columns. The plan resolves duplicates, fragile raw location text, heavy missingness in Eircode and Property Size Description, VAT-related price inconsistency, and extreme price outliers.

The overall principle is to keep interpretable business signal, replace unstable raw fields with compact engineered variables, and document each decision so the cleaning process remains auditable and reproducible.

2. DQP Scope and Baseline Input

2.1 Input Dataset for Cleaning

The executable DQP stage begins from geocoded_train.csv rather than the raw RPPR file because geocoding was performed offline. This separates a slow external-API step from the deterministic cleaning steps implemented in the notebook.

At load time, the baseline dataset contains 54,000 rows x 14 columns: the original RPPR fields plus geocode_status, latitude, longitude, geocode_provider, and dublin_district. The same two major missingness issues from DQR remain: Property Size Description is 94.75% missing and Eircode is 68.72% missing.

Baseline item	Observed value
Rows x columns	54,000 x 14
Duplicate rows	9
Property Size Description missing	51,164 (94.75%)
Eircode missing	37,108 (68.72%)
Geocoding columns	Only populated for Dublin rows by design

2.2 Cleaning Objectives

List every input feature, state its type, identify its data quality issue, and record the chosen cleaning action.

Fix the final analytical type for each retained variable and ensure no NaN values remain in the exported cleaned dataset.

3. Feature-level Decisions

Table 3.1 records the final decision for every feature present at the start of DQP. To keep the table readable in portrait layout, actions are written in compact single-line form and fuller justification is summarised in the subsection that follows.

Feature	Input type	Issue	Chosen action	Final type
Date of Sale	object	Raw text	Parse date; create Sale Year and Sale Month; drop raw date	Sale Year/Sale Month: Int64
Address	object	Near-unique free text	Create is_apartment; use offline geocoding support; drop raw field	is_apartment: int64
County	object	Too coarse for Dublin	Merge with dublin_district into location; drop source	location: category
Eircode	object	68.72% missing; high cardinality	Drop after location engineering	Dropped
Price (EUR)	object	Text, skew, outliers	Parse numeric; audit raw price; build Price_Log via VAT + clamp + log	Price: float64; Price_Log: float64
Not Full Market Price	object	Untyped binary text	Retain as categorical predictor	category
VAT Exclusive	object	Needed for price rule	Retain as categorical; use in VAT adjustment	category
Description of Property	object	Needed for VAT rule	Retain as categorical predictor	category
Property Size Description	object	94.75% missing	Drop; coverage too low for reliable use	Dropped
geocode_status	object	Workflow artifact	Drop after QA	Dropped
latitude	float/object	Auxiliary geocode field	Drop after district derivation	Dropped
longitude	float/object	Auxiliary geocode field	Drop after district derivation	Dropped
geocode_provider	object	Workflow artifact	Drop after QA	Dropped
dublin_district	object	Dublin-only helper field	Use to build location; drop source	location: category

3.1 Temporal field: Date of Sale

Issue: raw text and not model-ready. Options considered: keep raw, parse only, or derive temporal features. Chosen solution: parse with dayfirst=True, create Sale Year and Sale Month, and drop the original date because the derived fields preserve signal with less redundancy.

3.2 Raw location fields: Address, County, dublin_district, Eircode

Issue: Address and Eircode are high-cardinality, while County is too coarse for Dublin. Options considered: retain raw text, one-hot encode raw values, prefix-only Eircode use, or engineer compact geography. Chosen solution: extract is_apartment from Address, combine County with dublin_district into one location feature, then drop Address, Eircode, County, and dublin_district once their signal has been transferred.

3.3 Price and VAT-related fields

Issue: raw Price is text, highly skewed, and affected by a VAT-definition inconsistency for new dwellings. Options considered: model raw numeric price, log raw price only, or standardise the price definition first. Chosen solution: retain Price as an audit column, keep VAT Exclusive and Description of Property as predictors, apply a 13.5% VAT uplift where required, clamp extreme values using training-only percentile bounds, and use log1p of the final price as Price_Log.

3.4 Binary and sparse categorical fields

Issue: Not Full Market Price is untyped binary text; Property Size Description is overwhelmingly missing. Options considered: drop, impute, or retain selectively. Chosen solution: keep Not Full Market Price as category because it is complete and low-cardinality, but drop Property Size Description because 94.75% missingness makes imputation unreliable.

3.5 Geocoding auxiliaries

Issue: geocode_status, latitude, longitude, and geocode_provider describe the preprocessing workflow rather than the property itself. Options considered: keep for audit, use as predictors, or drop after QA. Chosen solution: drop them after district derivation because they are intermediate artifacts, not stable business features.

4. Implemented Cleaning Workflow and Outcomes

The implemented cleaning sequence is summarised in Table 4.1. No step introduces unresolved missingness, and no core validation rule rejects any row after parsing and field construction.

Step	Rows after	Cols after	Outcome
Load baseline	54,000	14	geocoded_train.csv loaded
Drop duplicates	53,991	14	9 exact duplicates removed
Create is_apartment	53,991	15	binary dwelling-type flag added
Drop Address + auxiliaries	53,991	10	intermediate columns removed
Basic standardization	53,991	12	typed fields and time features created
Construct location	53,991	11	51-category geography feature created
Core validation	53,991	11	0 rows rejected
VAT adjustment	53,991	13	9,513 rows adjusted
Price clamp + log	53,991	15	Price_Log created
Drop redundant columns	53,991	10	final compact schema reached
Final NaN check	53,991	10	0 missing values remain

5. Final Cleaned Dataset Specification

The exported DQP output contains one audit column, eight predictors, and one modelling target. All feature types are fixed and no missing values remain.

Retained column	Final type	Role	Reason for retention
Price (EUR)	float64	Audit	Declared price retained for back-checking and transparency.
Not Full Market Price	category	Predictor	Complete, low-cardinality business indicator.
VAT Exclusive	category	Predictor	Retained for interpretation and VAT standardisation.
Description of Property	category	Predictor	Captures dwelling class and supports the VAT rule.
is_apartment	int64	Predictor	Compact property-type signal from Address.
Sale Year	Int64	Predictor	Captures long-run market trend.
Sale Month	Int64	Predictor	Captures within-year timing.
location	category	Predictor	Unified geography with Dublin district granularity.
vat_adjusted_flag	int64	Predictor	Marks which rows were VAT-standardised.
Price_Log	float64	Target	Final modelling target: log1p of VAT-adjusted clamped price.

5.1 Final Validation Outcomes

Final shape: 53,991 rows x 10 columns.

Total remaining missing values: 0.

All Price (EUR) values are positive and all Price_Log values are valid for modelling.

Saved file: train-clean-final.csv.

6. Conclusion

This DQP satisfies the project requirement to identify feature-level issues, consider alternative solutions, justify the chosen actions, finalise feature types, and export a cleaned CSV with no missing values.

The most important choices are to replace fragile raw location text with compact engineered geography, standardise VAT-exclusive new-build prices before modelling, and control extreme target values through training-only clamping. Together these actions improve consistency, robustness, and interpretability.

Appendix A: Alignment with Part 2 of project.ipynb

§2.1: offline geocoding workflow documented and used as source of dublin_district.

§2.2: baseline check records shape, dtypes, missingness, and duplicate count.

§2.3: duplicate removal deletes 9 rows.

§2.4–§2.4.1: is_apartment extracted before Address and geocoding auxiliaries are dropped.

§2.5: standardisation performs whitespace cleaning, date/price parsing, temporal extraction, and category casting.

§2.6: County and dublin_district are merged into location.

§2.7: core validation rejects 0 rows.

§2.8: VAT adjustment affects 9,513 rows at rate 13.5%.

§2.9–§2.9.1: p05 = €60,000 and p95 = €721,137.50 are used for clamping; Price_Log is created.

§2.10–§2.13: sparse/redundant fields are dropped, final validation passes, and train-clean-final.csv is exported.